Disaggregating death rates of age-groups using deep learning algorithms.

EDSD workshop on Machine Learning

Andrea Nigri [1]

11 June, 2025

[1] Department of Economics, Management and Territory, University of Foggia

Article

Journal of Official Statistics
2024, Vol. 40(2) 262–282

DOI: 10.1177/0282423X241240739
journals.sagepub.com/home/jof
Sage

Disaggregating Death Rates of Age-Groups Using Deep Learning Algorithms

Andrea Nigri[1], Susanna Levantesi[2], and Salvatore Scognamiglio[3]

Abstract

Reliable estimates of age-specific vital rates are crucial in demographic studies, while ages are, in most cases, commonly grouped in bins of five years. Indeed, public health and national systems require single age-specific data to achieve accurate social planning. This paper introduces a deep learning approach for splitting the abridged death rates, providing a more comprehensive perspective on the indirect age-specific vital rates estimation from grouped data. Additionally, we contribute to the existing literature by introducing a multi-population (countries and genders) approach, providing reliable estimates considering the heterogeneity of longevity dynamics over age, years, and across populations. We also contribute to the state of the art in indirect estimation by introducing, for the first time, a multi-population indirect estimation leveraging subnational data. Our model accurately captures mortality dynamics by age over time and among different populations. We prove the model's ability to estimate reliable predictions of age-specific mortality rates by also studying how the hyperparameters' choice affects the model reliability and analyzing the age-specific relative differences between the real and the estimated mortality rates.

Keywords

mortality modeling, ungroup, deep learning

Outline

Outline

- Background & Motivation
- Ungrouping Estimation - State of the Art
- Methodological framework
- Application: Human Mortality Database (HMD)
- Conclusions

Background & Motivation

Steady improvement in mortality level

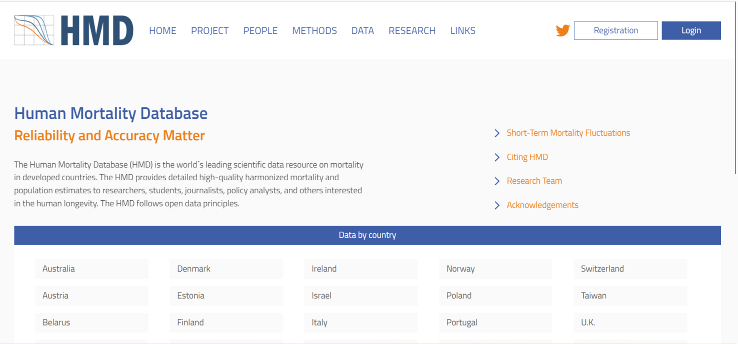
Reliable predictions

- Reliable predictions of age-specific death rates are crucial.
- **Issue:** Often ages are grouped in 5-year age classes (causes of death, COVID-19) - We need single ages
- **The most prominent framework** to ungroup mortality data: Composite link model (CLM) by Rizzi et al. 2015
- **Drawbacks:** Working on a single-gender population - lack of harmonization, comparability, and coherence among countries and gender.
- **Proposal:** A deep learning multi-population (countries and genders) approach for splitting the abridge death rates.

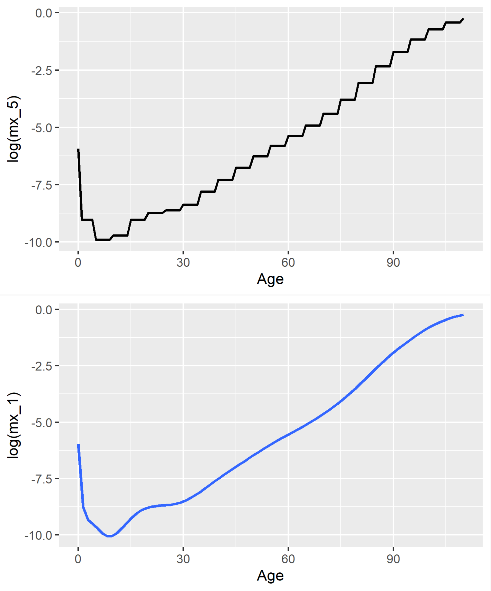
Data

- Data from the Human Mortality Database (HMD; mortality.org).
- 41 countries, both genders, ages and years

$$m_{x,t} = \frac{D_{x,t}}{E_{x,t}}$$



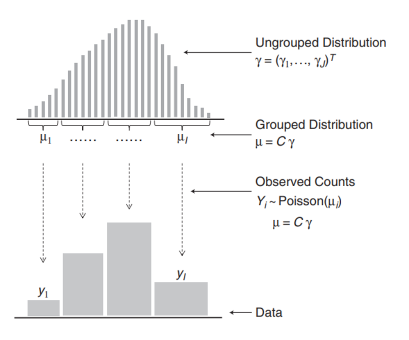
Basic Idea


0.0
-2.5
-5.0
-7.5
-10.0
log(mx_5)
0
30
60
90
Age
0.0
-2.5
-5.0
-7.5
-10.0
log(mx_1)
0
30
60
90
Age

State of the Art PCLM

PCLM

- The actually observed grouped counts y_i are realization from $Y_i \sim Pois(\mu_i)$, whose expected values $\boldsymbol{\mu} = E(Y) = C\gamma$, result from grouping the original distribution γ into bins
- C describes how the latent distribution was mixed before generating the data. C are 0 except for those $c_{ij} = 1$ that indicate the elements of γ that are aggregated into histogram bin i.
- This is a composite link model which extends standard generalized linear models. To guarantee nonnegative values of γ, we write it as $\gamma = e^{X\beta}$ and estimate the values of $\beta = (\beta_1, \ldots, \beta_J)^T$.



Contribution

The aim of this study is

- To formulate a multi population model leveraging on deep learning algorithms based on neural networks to derive death rates for single ages from 5y-grouped death rates
- The mortality pattern by age and time is indirectly identified by the network.
- Resulting estimates would be useful for guiding public health interventions, informing about age-specific mortlaity dynamics in contexts with deficient data collection.

Methodology: Fully-Connected Network (FCN) and the Embedding Network (EN)

Neural Network (NN)

$$y = \phi\left(w_0^{(o)} + \langle \boldsymbol{w}^{(o)}, \boldsymbol{z}^{(1)}(\boldsymbol{x})\rangle\right), \tag{1}$$

where $w_0^{(o)} \in \mathbb{R}$, $\boldsymbol{w}^{(o)} \in \mathbb{R}^{q_1}$, and $\langle \cdot, \cdot \rangle$ denotes the scalar product in $\mathbb{R}^{q_1}$.

Let $\boldsymbol{x} \in \mathbb{R}^{q_0}$ be the input vector; a FCN layer with $q_1 \in \mathbb{N}$ units is a vectorial function that maps $\boldsymbol{x}$ to a q_1-dimensional real-valued space:

$$\boldsymbol{z}^{(1)} : \mathbb{R}^{q_0} \to \mathbb{R}^{q_1}, \qquad \boldsymbol{x} \mapsto \boldsymbol{z}^{(1)}(\boldsymbol{x}) = \left(z_1^{(1)}(\boldsymbol{x}), z_2^{(1)}(\boldsymbol{x}), \ldots, z_{q_1}^{(1)}(\boldsymbol{x})\right)^\top. \tag{2}$$

The output of each unit is a new feature $\boldsymbol{z}^{(1)}(\boldsymbol{x})$ (in matrix form) which is a non-linear function of $\boldsymbol{x}$:

$$\boldsymbol{z}^{(1)}(\boldsymbol{x}) = \phi(\boldsymbol{w}_0^{(1)} + W^{(1)}\boldsymbol{x}). \tag{3}$$

where $\phi : \mathbb{R} \mapsto \mathbb{R}$ is the activation function, $w_{j,0}^{(1)}$ indicates the bias/intercept, and $w_{j,i}^{(1)} \in \mathbb{R}$ represent the weights.

In the case of deep NN, the output layer uses the features extracted by the last hidden layer $\boldsymbol{z}^{(h:1)}(\boldsymbol{x})$ instead of those $\boldsymbol{z}^{(1)}(\boldsymbol{x})$.

$$\boldsymbol{x} \mapsto \boldsymbol{z}^{(h:1)}(\boldsymbol{x}) = \left(\boldsymbol{z}^{(h)} \circ \cdots \circ \boldsymbol{z}^{(1)}\right)(\boldsymbol{x}) \in \mathbb{R}^{q_h}, \tag{4}$$

Embedding

- To extend the framework to the mp-DNN model by adding other demographic features, using Embedding tool.
- Embedding is a tool that allows for the capture of relationships that are very difficult to capture otherwise due to high dimensionality.
- Basically embedding allows a low-dimensional representation learning, mapping categorical variable into a vector space.
- Embedding layers map these all features into a real-valued $\mathbb{R}^{q_{\mathcal{L}}}$-dimensional space
- Formally, let $\mathcal{L} = \{l_1, l_2, \ldots, l_{n_{\mathcal{L}}}\}$ be the set of categories of the qualitative variable and $n_{\mathcal{L}}$ be its cardinality. An embedding layer is a mapping

$$\boldsymbol{z}_{\mathcal{L}} : \mathcal{L} \to \mathbb{R}^{q_{\mathcal{L}}}.$$

The number of embedding parameters to learn during the network calibration is $n_{\mathcal{L}} q_{\mathcal{L}}$.

DNN Optimization

- The NN training is generally carried out using the Gradient Descent algorithm or one of its extensions, where the updating of the weights is based on the gradient of the loss function $L(\boldsymbol{\theta})$.
- The weights are iteratively adjusted to decrease the error of the network outputs with respect to some reference values.
- Let $\boldsymbol{\theta}^{(t)}$ the vector of parameters at time t, the updating rule can be written as follows:

$$\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)} + \eta \nabla L(\boldsymbol{\theta})$$

  where $\eta \in [0, 1]$ is the learning rate, $\boldsymbol{\theta}^{(0)}$ is the initial weight configuration. We remark that the complexity of the training grows with the number of layers and units per layer in the network architecture.

The NN model for ungrouping mortality data

Ungrouping

We assume that the single-age death rate $\{m_{x,t,r,g}\}$ in log scale can be modelled as a function that depends on some inputs: the age x, region r, the gender g, the time t, and the death rate in log scale of the age-group $\nu(x)$ to which x belongs.

$$f : \mathcal{X} \times \mathcal{R} \times \mathcal{G} \times \mathcal{T} \times \mathbb{R} \to \mathbb{R}, \qquad \big(x, r, g, t, \log(m_{\nu(x),t,r,g})\big) \mapsto \log(m_{x,t,r,g}). \quad (5)$$

- f is unknown and potentially complex.
- we use deep NNs to find and approximation $\hat{f}(\cdot)$

We define the complete vector of features concatenating the embedding weights and the numerical inputs:

$$\boldsymbol{z}(x,t,r,g) = \big(\boldsymbol{z}_{\mathcal{X}}(x), t, \boldsymbol{z}_{\mathcal{R}}(r), \boldsymbol{z}_{\mathcal{G}}(g), \log(m_{\nu(x),t,r,g})\big)^{\top} \in \mathbb{R}^{q_{\mathcal{R}}+q_{\mathcal{G}}+q_{\mathcal{X}}+2}$$

$\boldsymbol{z}(x,t,r,g)$ is further processed with a three FCN layers of size $(q_1, q_2, 1) \in \mathbb{N}^3$.

Ungrouping

We design an architecture where the last layer of the network (output layer) has a size equal to 1 ($q_3 = 1$). To avoid overfitting, we introduce dropout among the FCN layers:

$$
\begin{aligned}
\boldsymbol{z}^{(1)}(x,t,r,g) &= \phi\Big(\boldsymbol{w}_0^{(1)} + W^{(1)}\boldsymbol{z}(x,t,r,g)\Big) \\
r_j^{(1)} &\sim \mathit{Bernoulli}(p_1) \\
\dot{\boldsymbol{z}}^{(1)}(x,t,r,g) &= \boldsymbol{r}^{(1)} * \boldsymbol{z}^{(1)}(x,t,r,g) \\
\boldsymbol{z}^{(2)}(x,t,r,g) &= \phi\Big(\boldsymbol{w}_0^{(2)} + W^{(2)}\dot{\boldsymbol{z}}^{(1)}(x,t,r,g)\Big) \\
r_j^{(2)} &\sim \mathit{Bernoulli}(p_2) \\
\dot{\boldsymbol{z}}^{(2)}(x,t,r,g) &= \boldsymbol{r}^{(2)} * \boldsymbol{z}^{(2)}(x,t,r,g) \\
\log(\hat{m}_{x,t,r,g}) &= \phi\Big(w_0^{(3)} + \boldsymbol{w}^{(3)}\dot{\boldsymbol{z}}^{(2)}(x,t,r,g)\Big),
\end{aligned}
$$

where $p_1, p_2 \in [0,1]$, and $\boldsymbol{w}_0^{(1)}, \boldsymbol{w}_0^{(2)}, w_0^{(3)}, W^{(1)}, W^{(2)}, \boldsymbol{w}^{(3)}$ are the weights.

Ungrouping

- The weight matrices and bias vectors of the FCN layers and the parameters of the embedding layers need to be appropriately calibrated.
- These parameters are iteratively adjusted via the BP algorithm to minimise a specific loss function.
- Following the practice adopted in the mortality literature, we fit our model using the Mean Squared Error (MSE). In such a case, the training of the network requires the minimisation of the loss:

$$L(\boldsymbol{\theta}) = \sum_{x \in \mathcal{X}} \sum_{t \in \mathcal{T}} \sum_{r \in \mathcal{R}} \sum_{g \in \mathcal{G}} \big(\log(m_{x,t,r,g}) - \log(\hat{m}_{x,t,r,g}) \big)^2.$$

where $\boldsymbol{\theta}$ denotes the NN parameters.

Implementation

Let (t_0, t_s), be the observed period,

Let $\left\{\log(m_{\nu(x),t,r,g})\right\}_{t=t_0}^{t_s}$, for $t_0 < t_s$, be the country-specific observed grouped death rates in log scale. Then, each series is split into a train-validation set and a test set, where the first is used for fitting the model's parameters, while the second is used to test the model's prediction and calculate the error.

- The best DNN setting during the training phase is used to obtain predictions in the test phase.
- Train -validation Input: $\left\{\log(m_{\nu(x),t,r,g})\right\}_{t=t_0}^{t_\tau}$,
- Train -validation Output: $\{\log(\hat{m}_{x,t,r,g})\}_{t=t_0}^{t_\tau}$.
- Test Input: $\left\{log(m_{\nu(x),t,r,g})\right\}_{t=t_\tau+1}^{t_s}$,
- Test Output: $\{\log(\hat{m}_{x,t,r,g})\}_{t=t_\tau+1}^{t_s}$.

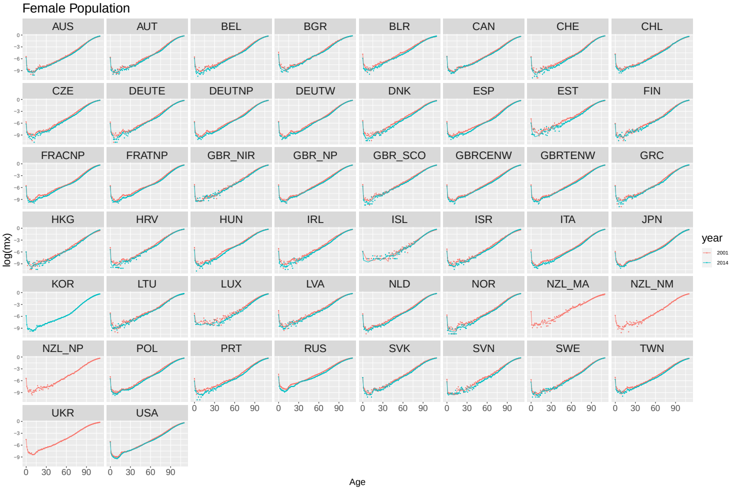
Thereby, denoting $\hat{f}_{nn}$ as a composition of functions defined based on the NN architecture, the model can be described by:

$$\{\log(\hat{m}_{x,t,r,g})\}_{t=t_\tau+1}^{t_s} = \hat{f}_{nn}\left(\left\{\log(m_{\nu(x),t,r,g})\right\}_{t=t_\tau+1}^{t_s}\middle|\hat{\boldsymbol{\theta}}\right), \tag{6}$$

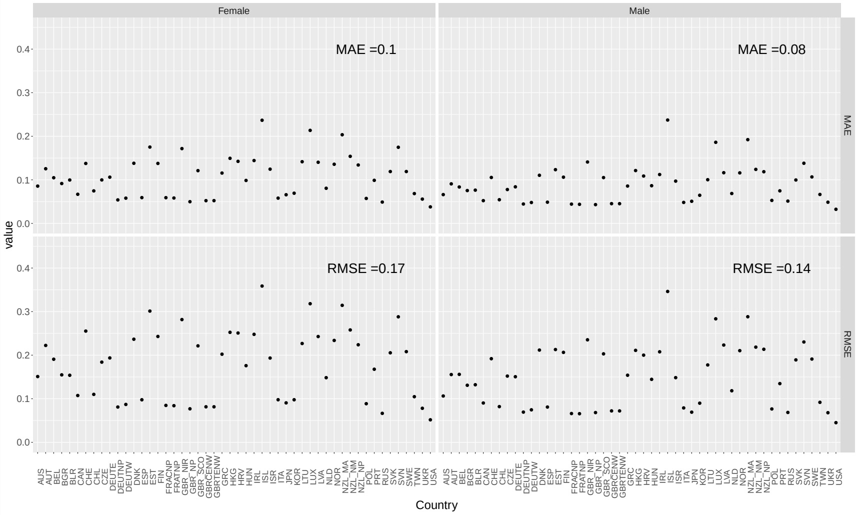
where $\{\log(\hat{m}_{x,t,r,g})\}_{t=t_\tau+1}^{t_s}$ are the death rates in log scale in single year age-group, in the test set obtained by $\hat{f}_{nn}$, that involves the NN weights $\hat{\boldsymbol{\theta}}$ estimated during the network training.

Results

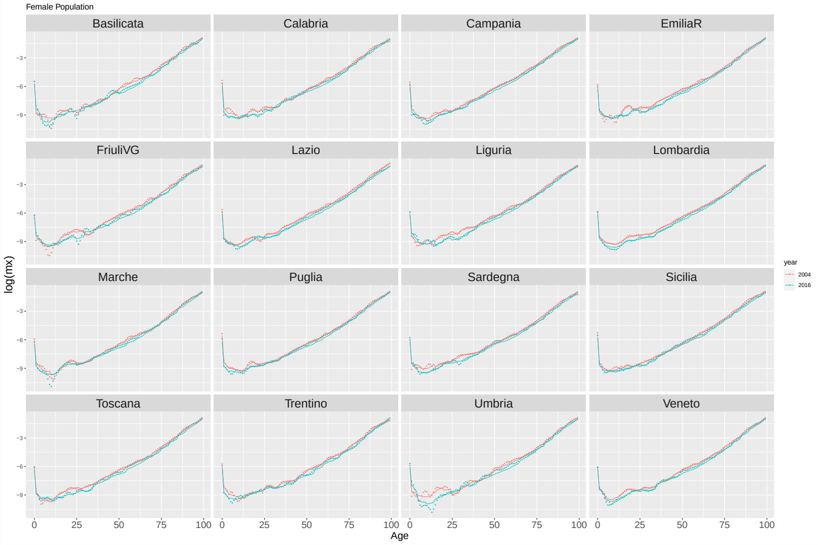
- We consider historical mortality data for all available countries and both genders. Aiming to assess the multi population model robustness and consistency toward the historical data.
- 30 years for training-validation (1970-2000) and the remaining years (2001-2015) are used for model forecasting/test, i.e. to test the model's prediction and calculate the error.
- We validate the DNN model performance investigating whether the approaches can capture (a) regular and irregular trends over time (b) dynamics of age-specific mortality improvements.
- Analysis includes numerical and graphical representations of the goodness of fit RMSE and MAE on the out-of-sample period 2001-2015 time window
- Appendix: Parameters and structural reliability analysis of DNN



Estimated ungrouped death rates, years 2001 and 2014 based on the period 1970-2000 (training and validation). Females.


Female
Male
MAE =0.1
MAE =0.08
RMSE =0.17
RMSE =0.14
MAE
RMSE
value
Country
AUS AUT BEL BGR BLR CAN CHE CHL CZE DEUTE DEUTNP DEUTW DNK ESP EST FIN FRACNP FRATNP GBR_NIR GBR_NP GBR_SCO GBRCENW GBRTENW GRC HKG HRV HUN IRL ISL ISR ITA JPN KOR LTU LUX LVA NLD NOR NZL_MA NZL_NM NZL_NP POL PRT RUS SVK SVN SWE TWN UKR USA

Subregional data



Conclusions

Conclusions

- This paper contributes to the current literature on the demographic methods for ungrouping vital rates leveraging deep learning,
- We propose a DNN model in a multi-population framework to ungroup death rates from rates gathered in five-year age-groups
- The results of the numerical experiments show the high accuracy of the proposed model, which captures the dynamics of mortality by age over time and between different populations.
- We also contribute to the state of the art in indirect estimation by introducing a multi-population indirect estimation leveraging subnational data.

Thank you for your attention.

Questions / Suggestions ?

Appendix

Appendix: Parameters and structural reliability analysis of DNN

Layer	Error	Relu		Sigmoid		Softmax	
		Drop-out	No Drop-out	Drop-out	No Drop-out	Drop-out	No Drop-out
1	MAE	0.09	0.10	0.11	0.10	0.16	0.11
	RMSE	0.15	0.16	0.17	0.16	0.23	0.18
2	MAE	0.10	0.09	0.09	**0.08**	0.15	0.11
	RMSE	0.16	0.15	0.15	**0.14**	0.21	0.18
3	MAE	0.14	0.10	0.10	0.09	2.32	2.33
	RMSE	0.19	0.16	0.16	0.16	2.69	2.69